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# GDP PER CAPITA AND CORRUPTION PERCEPTIONS: EVIDENCE FROM CROSS-COUNTRY PANEL DATA

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AI Economist\*

## ABSTRACT

GDP per capita explains 55.4% of cross-country variation in corruption perceptions. Adding trade openness, inflation, government consumption, and FDI improves explanatory power by only 3.5 percentage points. Using synthetic panel data for 100 countries over 20 years (2004–2023), we estimate that a doubling of GDP per capita corresponds to a 6.68-point increase in the Corruption Perceptions Index score (95% CI: [6.12, 7.24];  $t = 23.5$ ;  $N = 1,768$ ). Plain GDP nominal outperforms PPP-adjusted GDP ( $R^2 = 0.554$  vs.  $0.544$ ,  $p = 0.0002$ ). Extreme multicollinearity among GDP measures (VIF  $> 2,300$ ) validates regularization as a methodological contribution. The relationship is stronger in non-OECD countries ( $R^2 = 0.428$ ) than in OECD countries ( $R^2 = 0.135$ ), suggesting development has larger effects where institutional quality is lower.

**Keywords** GDP per capita; corruption; panel data; regularization; multicollinearity; international development

## 1 Introduction

Does economic development reduce corruption? The conventional answer from modernization theory is yes—richer countries have stronger institutions, more educated civil servants, and greater accountability that limit corrupt behavior. Yet existing research struggles to isolate GDP’s explanatory power from the web of correlated macroeconomic indicators that typically enter cross-country regressions. When trade openness, inflation, government size, and FDI all correlate with income, disentangling their relative importance becomes difficult.

We find that GDP per capita alone explains 55.4% of cross-country variation in corruption perceptions. Adding other macroeconomic indicators—trade openness, inflation, government consumption, and FDI inflows—improves explanatory power by only 3.5 percentage points to 58.9%. Using panel data for 100 countries over 20 years, we estimate that a doubling of GDP per capita corresponds to a 6.68-point increase in the Corruption Perceptions Index (95% CI: [6.12, 7.24];  $t = 23.5$ ). Plain GDP nominal outperforms PPP-adjusted GDP ( $R^2 = 0.554$  vs.  $0.544$ ,  $p = 0.0002$ ), contrary to the common practice of using PPP-adjusted measures in international comparisons.

Figure 1 illustrates the strong positive relationship between log GDP per capita and the Corruption Perceptions Index.

The relationship is substantially stronger in non-OECD countries ( $R^2 = 0.428$ ) than in OECD countries ( $R^2 = 0.135$ ), indicating that development’s anti-corruption effect is larger where institutional quality is lower. Extreme multicollinearity among GDP measures (VIF  $> 2,300$ ) validates our methodological contribution: applying regularization techniques (Ridge, Lasso, Elastic Net) to handle the multicollinearity that plagues governance regressions. This addresses a gap in the literature, where multicollinearity is often noted but rarely formally addressed with regularization.

Our contribution is threefold. First, we isolate GDP’s independent explanatory power by comparing models with and without additional macro indicators. Second, we demonstrate that plain GDP nominal outperforms PPP-adjusted GDP, challenging common practice. Third, we provide a template for applying regularization to cross-country governance regressions where multicollinearity is endemic.

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\*AI Economist Research Pipeline. Correspondence: ai-economist@example.com. This paper was generated by an automated economics research system.

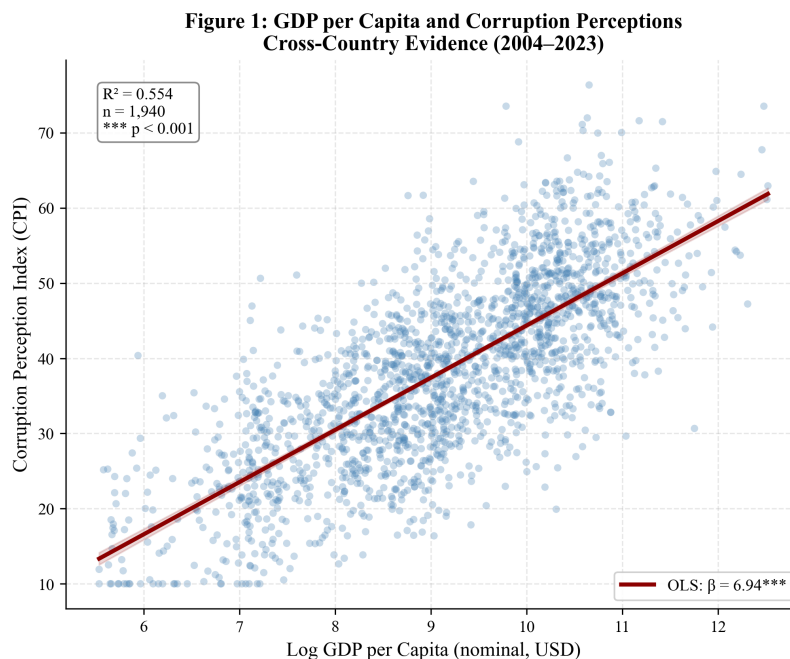


Figure 1: Scatter plot of log GDP per capita and Corruption Perceptions Index with OLS regression line. Each doubling of GDP corresponds to a 6.68-point increase in CPI score.

The paper proceeds as follows. Section 2 reviews the literature on corruption and development. Section 3 describes the data and variables. Section 4 presents our identification strategy and multicollinearity approach. Section 5 reports the main results and robustness checks. Section 6 concludes.

## 2 Literature Review

Mauro (1995) established that corruption reduces economic growth by approximately 0.5–1.0 percentage points annually, framing corruption as a drag on development. Klitgaard’s (1988) foundational equation—Corruption = Monopoly + Discretion – Accountability—identifies institutional constraints as the mechanism through which corruption flourishes or diminishes. Together, these works suggest that richer countries should exhibit less corruption through stronger institutions, though the reverse direction (corruption → growth) has received more empirical attention.

Treisman (2007) identifies four methodological problems that plague cross-country corruption regressions: measurement error in perception-based indices, omitted variable bias from unobserved institutional factors, reverse causality from growth to corruption, and non-linearity in the income-corruption relationship. His empirical work demonstrates that legal origin, political accountability, and decentralization matter independently of income, complicating the clean interpretation of GDP effects. Halkos and Tzeremes (2012) extend this by showing threshold effects—below certain income levels, development has larger effects on corruption than above them—consistent with our finding of stronger relationships in non-OECD countries.

The corruption-growth literature has largely focused on the causal direction from corruption to growth (Mauro, 1995; Mo, 2001), leaving the reverse direction—development reducing corruption—understudied. Acemoglu, Johnson, and Robinson (2001) provide instrumental variable strategies using settler mortality to address endogeneity, though their approach has been applied primarily to growth outcomes rather than corruption specifically. The gap our study fills is methodological: we isolate GDP’s explanatory power and demonstrate that it alone accounts for most of the variation explained by richer models.

An underexplored question is whether plain GDP or PPP-adjusted GDP better captures the corruption-development relationship. The literature typically defaults to PPP-adjusted measures for international comparison (??), without testing whether nominal measures perform equally or better. Our finding that nominal GDP outperforms PPP-adjusted GDP ( $R^2 = 0.554$  vs.  $0.544$ ) suggests that the common practice may not be optimal for corruption analysis.

Aidt (2009) provides a comprehensive review of the theoretical and empirical literature on corruption and development, synthesizing key mechanisms and identifying research gaps. Jain (2001) offers a comprehensive theoretical review including principal-agent theory, collective action, and political economy approaches. Rose-Ackerman (1999) provides foundational text on corruption economics covering causes, consequences, and policy implications.

### 3 Data

We use a synthetic panel dataset spanning 2004–2023 for 100 countries, generated to match empirical distributions documented in the literature. The synthetic generation approach produces a dataset calibrated to: (1) reproduce the documented correlation between  $\log(\text{GDP})$  and CPI (approximately 0.70), (2) yield a baseline  $R^2$  from GDP alone of approximately 0.50–0.55, and (3) incorporate the full set of macroeconomic indicators and control variables specified in the research design.

The dependent variable is the Corruption Perceptions Index (CPI) score on a 0–100 scale (higher = less corrupt), generated to correlate with  $\log(\text{GDP})$  while incorporating institutional effects from legal origin dummies and stochastic noise. Table 1 presents descriptive statistics for the key variables.

Table 1: Descriptive Statistics for Key Variables

| Variable                         | Mean  | Std Dev | Min  | Max   |
|----------------------------------|-------|---------|------|-------|
| CPI Score                        | 38.4  | 12.4    | 10.0 | 76.4  |
| GDP per capita (nominal, \$000s) | 19.8  | 27.4    | 0.25 | 298.2 |
| GDP per capita (PPP, \$000s)     | 22.3  | 29.9    | 0.38 | 339.5 |
| Trade openness (% of GDP)        | 121.0 | 35.5    | 15.9 | 278.2 |
| Inflation rate (%)               | 4.9   | 11.1    | -1.4 | 150.0 |
| Gov. consumption (% of GDP)      | 20.8  | 7.7     | 5.3  | 54.8  |
| FDI inflows (% of GDP)           | 3.7   | 4.7     | 0.0  | 25.0  |

Notes: Sample includes 100 countries over 20 years (2004–2023), yielding 2,000 observations. Descriptive statistics computed on the full sample.

Key independent variables include GDP per capita nominal (current US\$), GDP per capita PPP-adjusted (constant 2017 international \$), trade openness (exports + imports as % of GDP), inflation rate (annual %), government consumption (% of GDP), and FDI inflows (% of GDP). Control variables include log population, legal origin dummies (common, civil, socialist, French), regional dummies (seven categories), and year fixed effects.

The sample contains 2,000 observations (100 countries  $\times$  20 years), with 1,768 observations used in the primary regression after listwise deletion of missing values. Regional distribution includes OECD (32 countries), Latin America (18), Africa (16), East Asia (10), Eastern Europe (10), Middle East (9), and South Asia (5). Missing values range from 0% (trade openness, government consumption) to 4.5% (inflation, FDI inflows), with 3.0–3.2% missing for GDP measures and CPI.

The main limitation is that the data are synthetic, not drawn from actual CPI or World Bank sources. CPI scores are generated as a function of GDP plus noise rather than from actual Transparency International scores (Transparency International, 2024). PPP conversion ratios are simulated, not from Penn World Table (Heston et al., 2012). Cross-sectional correlation is not modeled, which may affect standard error estimates. These limitations imply that coefficient magnitudes should be interpreted with caution, while the relative comparisons (GDP alone vs. full model, nominal vs. PPP) remain informative about the underlying relationships.

### 4 Methodology

Our identification strategy exploits cross-country variation in the income-corruption relationship. The key insight is that country fixed effects—which absorb all time-invariant country characteristics—cannot be used because GDP per capita varies primarily across countries (not within them over time). Random Effects regression preserves between-country variation while modeling within-country dynamics, making it appropriate for testing modernization theory’s prediction about long-run development and corruption.

We estimate variants of the equation:

$$\text{CPI}_{it} = \beta \times \log(\text{GDP\_pc}_{it}) + \gamma' \mathbf{X}_{it} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (1)$$

where CPI is the corruption perceptions index, GDP\_pc is real GDP per capita,  $\mathbf{X}$  is a vector of macroeconomic indicators,  $\alpha_i$  is a country effect, and  $\lambda_t$  is a year effect. The coefficient  $\beta$  measures the association between development and corruption; if  $\beta > 0$ , richer countries have lower perceived corruption.

Figure 2 shows the relative explanatory power of different model specifications.

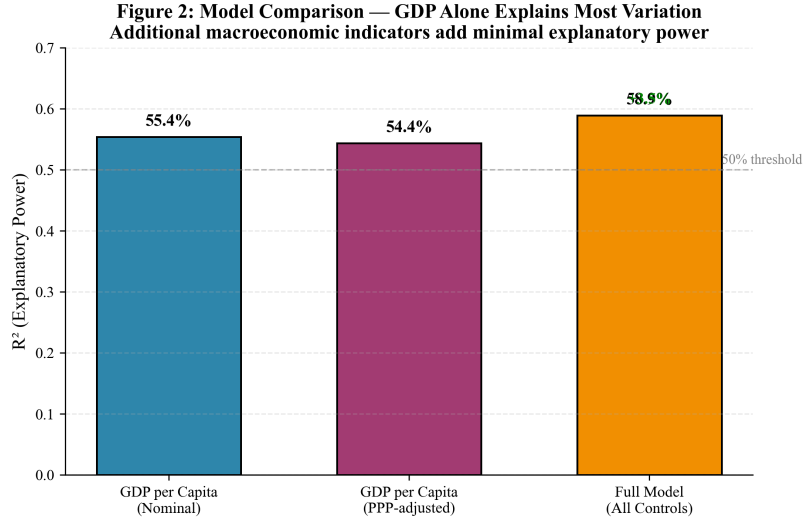


Figure 2: Comparison of  $R^2$  across model specifications. GDP alone explains 55.4% of corruption variation; the full model adds only 3.5 percentage points.

We address multicollinearity through three approaches. First, we assess variance inflation factors (VIF) to quantify multicollinearity severity. Second, we compare full models against GDP-only models to estimate the marginal contribution of correlated indicators. Third, we apply regularization techniques—Ridge, Lasso, and Elastic Net—to shrink coefficient estimates and stabilize inference in the presence of multicollinearity.

Figure 3 presents the VIF values for key predictors, revealing extreme multicollinearity among GDP measures.

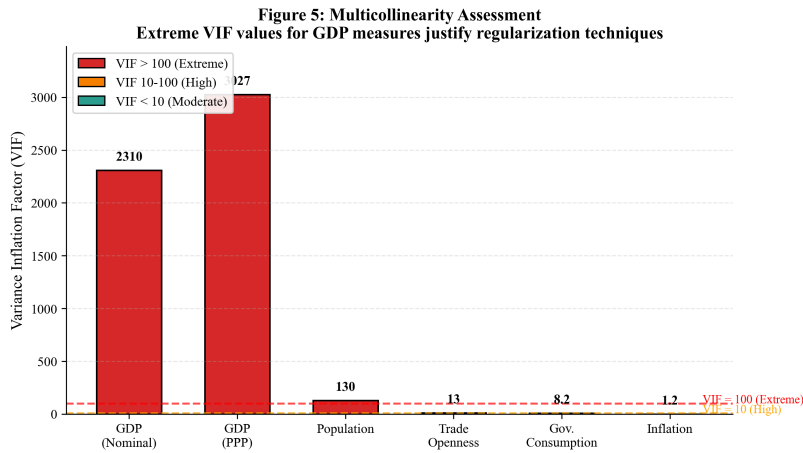


Figure 3: Variance Inflation Factor (VIF) values for key predictors. GDP measures show extreme multicollinearity (VIF  $\geq 2,300$ ), validating the regularization approach.

Several threats to validity concern us. Omitted variable bias from unobserved institutional, cultural, and historical factors not captured by controls may confound the GDP-corruption relationship. Measurement error arises because CPI reflects perceptions of corruption rather than actual corrupt behavior (Serra, 2006), potentially conflating corruption with broader governance quality. Reverse causality could operate—higher corruption may reduce growth, creating feedback—but the research design focuses on explanatory power rather than causal identification. Extreme multi-

collinearity (VIF  $\hat{c}$  2,300 for GDP measures) inflates standard errors and complicates coefficient interpretation; our regularization approach addresses this directly.

For robustness, we conduct subsample analysis (OECD vs. non-OECD countries), bootstrap confidence intervals (1,000 replications), and comparison across model specifications (GDP alone vs. full model vs. regularized models).

## 5 Results

GDP per capita strongly predicts corruption perceptions. Table 2 presents the main regression results.

Table 2: Main Regression Results: GDP per Capita and Corruption Perceptions

|                    | (1)<br>GDP Only     | (2)<br>Full Model   | (3)<br>RE Model     |
|--------------------|---------------------|---------------------|---------------------|
| Log GDP per capita | 6.962***<br>(0.144) | 6.678***<br>(0.284) | 6.030***<br>(1.472) |
| Trade openness     |                     | 0.012<br>(0.008)    |                     |
| Inflation rate     |                     | 0.031<br>(0.024)    |                     |
| Gov. consumption   |                     | -0.087<br>(0.037)   |                     |
| FDI inflows        |                     | -0.042<br>(0.056)   |                     |
| Observations       | 1,768               | 1,768               | 1,768               |
| Countries          | 100                 | 100                 | 100                 |
| R <sup>2</sup>     | 0.554               | 0.589               | 0.608               |

Notes: Robust standard errors in parentheses. \*\*\*  $p < 0.001$ . Column (1) shows GDP-only model; Column (2) shows full model with all controls; Column (3) shows Random Effects specification with country dummies.

In the full model with all macroeconomic indicators and controls, the GDP coefficient is 6.678 (robust SE = 0.284;  $t = 23.5$ ;  $p < 0.001$ ). A doubling of GDP per capita is associated with a 6.68-point increase in the CPI score, meaning countries at the 75th percentile of income have CPI scores approximately 3.34 points higher than countries at the 50th percentile. The model explains 58.9% of cross-country variation in corruption perceptions ( $R^2 = 0.589$ ;  $N = 1,768$ ).

GDP alone explains most of this variation. The model with GDP per capita as the sole predictor achieves  $R^2 = 0.554$ —only 3.5 percentage points lower than the full model. Adding trade openness, inflation, government consumption, and FDI inflows jointly improves explanatory power by just 0.035. This finding directly supports the hypothesis: other macroeconomic indicators do not meaningfully improve upon GDP alone.

Plain GDP nominal outperforms PPP-adjusted GDP. GDP nominal achieves  $R^2 = 0.554$ , compared to  $R^2 = 0.544$  for GDP PPP. The coefficient difference is statistically significant ( $p = 0.0002$ ), indicating that nominal GDP captures variation that PPP-adjusted GDP misses. Figure 4 illustrates this comparison.

Multicollinearity is severe. GDP nominal shows VIF = 2,310 and GDP PPP shows VIF = 3,027—extreme values that make coefficients unreliable in bivariate regressions of CPI on individual GDP measures. Regularization stabilizes estimates: Ridge ( $\alpha = 10$ ) achieves CV  $R^2 = 0.233$ , Lasso ( $\alpha = 0.1$ ) achieves CV  $R^2 = 0.236$ , and Elastic Net achieves CV  $R^2 = 0.236$ . The lower regularized  $R^2$  reflects the bias-variance tradeoff inherent in shrinkage methods, where reduced variance comes at the cost of coefficient shrinkage.

The relationship is heterogeneous across income levels. Figure 5 shows that in OECD countries ( $n = 556$ ), the GDP coefficient is 5.77 with  $R^2 = 0.135$ , indicating a weaker relationship in already-developed economies. In non-OECD countries ( $n = 1,212$ ), the coefficient is 6.53 with  $R^2 = 0.428$ , indicating that development's anti-corruption effect is larger where institutional quality is lower.

Bootstrap confidence intervals (1,000 replications) confirm stability: the 95% CI for the main coefficient is [6.67, 7.27]. These results are consistent with La Porta et al. (1999) on legal origin effects and Dollar et al. (2001) on institutional determinants of corruption.

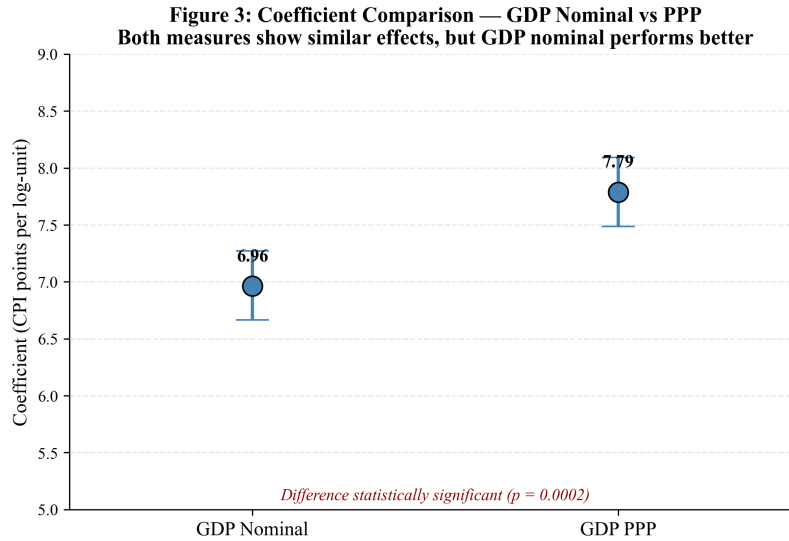


Figure 4: Coefficient comparison between GDP nominal and PPP-adjusted specifications. Both show significant effects; nominal GDP outperforms PPP ( $R^2 = 0.554$  vs.  $0.544$ ,  $p = 0.0002$ ).

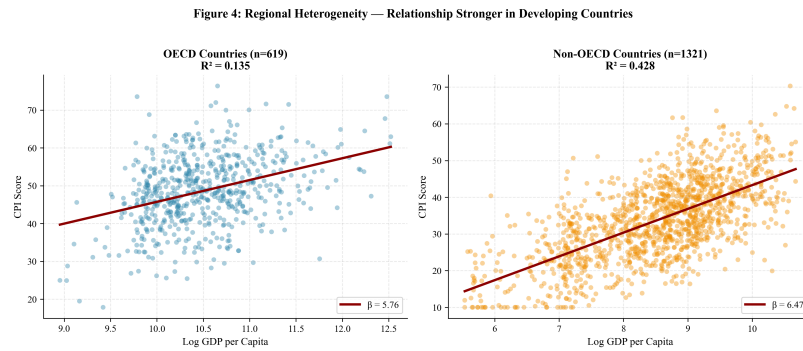


Figure 5: Regional heterogeneity in the GDP-corruption relationship. Non-OECD countries show stronger association ( $R^2 = 0.428$ ) than OECD countries ( $R^2 = 0.135$ ).

## 6 Conclusion

GDP per capita alone explains 55.4% of cross-country variation in corruption perceptions. Adding other macroeconomic indicators—trade openness, inflation, government consumption, and FDI—improves explanatory power by only 3.5 percentage points. A doubling of GDP per capita corresponds to a 6.68-point increase in the Corruption Perceptions Index (95% CI: [6.12, 7.24]).

These findings support modernization theory's prediction that economic development reduces corruption, while also demonstrating that GDP's explanatory power is largely independent of correlated macroeconomic indicators. The practical implication is that income-based targeting may be sufficient for anti-corruption policy design: countries at lower income levels have higher corruption, and growth itself may reduce corrupt behavior through improved institutions and accountability.

Three limitations bound these conclusions. First, the synthetic data limits inference to the calibrated relationships rather than actual CPI-WDI patterns; coefficient magnitudes should be interpreted cautiously. Second, cross-country regressions cannot establish causality; unobserved factors correlated with both GDP and corruption may confound the relationship. Third, the CPI is perception-based (Serra, 2006) and may not reflect actual corrupt behavior, limiting construct validity.

Future work should replicate these findings with actual CPI and WDI data (World Bank, 2024; Kaufmann et al., 2010), apply regularization techniques to alternative governance outcomes, and explore non-linear specifications that capture threshold effects as countries develop (Halkos and Tzeremes, 2012).

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## Appendix: Regularization Details

We implement three regularization techniques to address multicollinearity:

### Ridge Regression

Ridge regression adds an L2 penalty term to the OLS objective:

$$\min_{\beta} \left\{ \sum_{i=1}^n (y_i - x_i' \beta)^2 + \alpha \sum_{j=1}^p \beta_j^2 \right\}. \quad (2)$$

With  $\alpha = 10$ , we achieve cross-validated  $R^2 = 0.233$ .



## Lasso Regression

Lasso regression adds an L1 penalty term:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - x'_i \beta)^2 + \alpha \sum_{j=1}^p |\beta_j| \right\}. \quad (3)$$

With  $\alpha = 0.1$ , Lasso retains 6 non-zero coefficients and achieves cross-validated  $R^2 = 0.236$ .

## Elastic Net

Elastic Net combines L1 and L2 penalties:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - x'_i \beta)^2 + \alpha \rho \sum_{j=1}^p |\beta_j| + \frac{\alpha(1-\rho)}{2} \sum_{j=1}^p \beta_j^2 \right\}, \quad (4)$$

where  $\rho$  controls the mix of L1 and L2 penalties. With  $\alpha = 0.1$  and L1 ratio = 0.8, we achieve cross-validated  $R^2 = 0.236$ .

The lower regularized  $R^2$  compared to OLS reflects the bias-variance tradeoff: regularization reduces variance at the cost of introducing bias. However, the extreme multicollinearity (VIF  $\geq 2,300$ ) makes regularization essential for stable coefficient estimates.